

3(a)	m is constant <i>or</i> k/m is constant <u>and</u> so acceleration / a proportional to displacement / x	B1
	negative sign shows that acceleration / a is in opposite direction to displacement / x <i>or</i> negative sign shows acceleration / a is towards fixed point	B1
3(b)	evidence of comparison to expression to $a = -\omega^2 x$	B1
	$\omega^2 = k/m$ <i>or</i> $\omega^2 = 4.0/m$ hence $\omega = 2.0/\sqrt{m}$	A1
3(c)	$E_K = \frac{1}{2} m \omega^2 x_0^2$ <i>or</i> $E_K = \frac{1}{2} m v^2$ <u>and</u> $v = \omega x_0$	C1
	$= \frac{1}{2} m (4.0/m) (3.0 \times 10^{-2})^2$	C1
	$= 1.8 \times 10^{-3} \text{ J}$	A1

Question	Answer	Marks
3(d)	new $x_0 = \sqrt{[(1.8 \times 10^{-3} / 2) \times (2 / m \times (m / 4.0))]}$ or ($E_K \propto x_0^2$ so) new $x_0 = \sqrt{[\frac{1}{2} \times (3.0 \times 10^{-2})^2]}$	C1
	$= 2.12 \times 10^{-2} \text{ m}$	A1
3(e)	flux linked to block changes / flux is cut by block which induces an e.m.f. in block	B1
	(eddy) currents induced in block cause heating	B1
	thermal / heat energy comes from (kinetic / potential) energy of oscillations / block	B1

Question	Answer	Marks
4	piezo-electric / quartz crystal / transducer	B1
	<u>alternating</u> p.d. applied across crystal / transducer	B1
	causes crystal to vibrate / resonate	B1
	crystal resonates at ultrasound frequencies / crystal's natural frequency is in the ultrasound range / alternating p.d. is in ultrasound frequency range	B1